This supplement supplies the construction that sec. 13.7 defers here: exactly how the moving-sentinel world is built, how the three actions and the expected-free-energy policy move an agent, and how the *isolated*, *communicating*, and *EFE-guided* conditions differ. It answers the mechanical question left open in the main section — what precisely is held fixed and what varies across the three conditions whose accuracies are contrasted there — so the reported binary-complement numbers can be read against their generative model rather than taken on trust.

The moving-world generative model is built by `build_moving_world`. A linear grid of 4 cells holds one binary hidden state — the half of the grid (left = state 0, right = state 1) that contains the threat. The 2 sentinels start at evenly tiled positions $(i \cdot \lfloor n_{\text{positions}}/n_{\text{agents}} \rfloor)$ and each observe a half-open field-of-view window. With the default setup the two FOVs are disjoint, one per half. Each agent's likelihood is a 2×2 matrix over outcomes (detected / not_detected) given the binary state, with a confident signed reading for the half the agent watches; the transition tensor encodes three deterministic control paths — stay, left (reflecting at cell 0), and right (reflecting at the last cell). The hidden-state prior is uniform.

Supplement: moving-world methods and condition definitions I

Action selection has two regimes. The random conditions draw each agent's move uniformly from the three controls. The EFE-guided condition uses `efe_policy_select`: for every candidate move it lands the agent at the deterministic next position, reconstructs the likelihood from that viewpoint, and scores the move by the expected posterior entropy after one observation, $H = \sum_o P(o)\, H(P(s \mid o))$ — taking the entropy-minimizing (information-seeking) step.

Supplement: moving-world methods and condition definitions I

We compare three conditions — *isolated* (random moves, no sharing), *communicating* (random moves plus a per-step log-linear-pool consensus), and *EFE-guided* (information-seeking moves plus the same per-step sharing) — over 960 trials of 6 steps each, scoring the pooled consensus against the true state. All numerics are deterministic given the run seed.